

Introduction

Modern age electric vehicles (EVs), since their conceptualization in the early 19th century, have evolved through a series of historical events and technological advancements, gaining mass popularity among personal vehicle users in the US. In the early times, the fundamental benefits EVs offered over steam, fuel, and gasoline-powered engines included a noiseless experience, ease in driving, and no emission of smelly pollutants. Its growing popularity gradually accelerated in multiple phases, starting with the growing access of people to electricity in the early 20th century. Immediately after the 1973 Arab Oil Embargo, its popularity experienced a temporary, but sharp, spike as it offered an affordable alternative amidst soaring oil prices and gasoline scarcity. Soon, it regained its market when environmental concerns were prioritized in state and federal regulations. For example, the 1990 Clean Air Act Amendment, as well as the 1992 Energy Policy Act, along with some state-level transportation emission regulations, helped the mass interest in EVs regrow (US DoE, n.d.). Later, the 21st century, so far, has been a golden era for developments in the EV industry. Apart from the technological advancements to enhance the efficiency of the cars, manufacturers focused on making cars safer and futuristic. This, alongside addressing the ecological impacts of transportation-related emissions, is making positive contributions to sustainable transportation development by promoting public health and safety.

However, like any other systems or interventions, EVs also have some challenges. The anxiety about running out of battery power before reaching their destination, also known as range anxiety, is one of them (Pevec, 2020). In addition to vehicle price and charging availability, range anxiety is one of the most significant barriers to EV adoption, particularly among fleet operators (GEOTAB, 2025). To facilitate long-distance travel in an environmentally friendly and safe way using EVs, while attracting more users to EVs, the infrastructural gap fostering range anxiety needs to be addressed.

This project aims to explore the trend of the adoption of EVs and regional variation to support my MS thesis. Two research questions have been developed to guide my work as follows:

- How are EV charging stations spatially distributed? Is there any disproportionate distribution in charger type, accessibility?
- Is there any regional variation in the spatial distribution patterns? If so, what could be the key geospatial determinants of these patterns? (e.g., demography, land cover, road density, etc.)

Literature Review

EVs are at a disadvantage compared to conventional vehicles in terms of purchase cost (Yong et al., 2015), due to complex technology, expensive batteries, a smaller market, and the use of rare earth elements like dysprosium, terbium, praseodymium, and neodymium for motors, along with materials extracted from the earth (Ballinger et al., 2019; Garratt, 2024). Battery technology remains the weakest aspect of EV development, with low energy density limiting all-electric range and long-distance travel. Although investments have been made, the complexity of battery tech means progress has been slow. Range and performance remain issues despite advancements (Ehsani et al., 2018).

Among chargers, DCFC is fastest but also costliest and can degrade battery life due to heat, requiring quicker battery replacement (Javed, 2023). Charging costs vary by location, battery size, vehicle range, and state electricity rates (Qmerit, 2024). Efficient EV use depends on infrastructure, or EVSE. Barriers like poor battery tech, insufficient and poorly placed charging stations, and uncoordinated charging limit EV adoption (Mukherjee & Gupta, 2015; Sortomme & El-Sharkawi, 2011). Properly placed charging stations can address range anxiety (Deb et al., 2018).

Other infrastructure issues include permitting, space limitations, lack of billing models, grid stress, security, legal constraints, opposition from fossil fuel interests, pricing inconsistencies, and low public awareness (Meyer, 2017; The Times of India, 2017; RMI, 2018; Deb et al., 2018). In the US, raising funds is difficult due to regulatory misunderstandings of the cost–benefit analysis (RMI, 2018). Researchers have tackled EVCS placement using various models and algorithms (Ahmad, 2022). Studies in cities like Lodz, Dublin, Salt Lake City, Ankara, Uppsala, and across the Netherlands have used ArcGIS and QGIS tools to optimize EVCS locations (Soczówka et al., 2024; Charly et al., 2023; Shepero & Mukhammar, 2018; Erbas et al., 2018; Zhang & Iman, 2017).

Methodology

Study Area

As this study aims to analyze the distribution pattern of electric vehicle (EV) charging stations in the United States and evaluate their role in facilitating long-distance travel, it will include all charging stations across the United States. Also, the variation will be examined in the US State-wise. For this research work, the United States, consisting of 48 contiguous states, will be my study area. I will not include Alaska and Hawaii.

Data and sources

This study is mainly secondary data source-based research, so no survey will be required. All the data will be collected from reliable secondary data sources. To carry out the research project, the following list of data has been finalized considering the associated research questions.

Table 1: List of Data and Sources

Questions	List of data	Sources
1. How are EV charging stations spatially distributed? Is there any disproportionate distribution in charger type, accessibility?	• List of stations & geospatial location with types • Vehicle Registration • Administrative boundary map	• US Department of Energy • Bureau of Transportation Statistics • USGS National Map

2. Is there any regional variation in the spatial distribution patterns? If so, what are the key geospatial determinants of these patterns? (e.g., land cover, road density, demography, etc.)	• EV charging station locations • Road Network map • Administrative boundary map • Land cover map • Population density data/ EV users' data • State-wise GDP	• US Department of Energy • USGS National Map • USGS National Land Cover Database • US Census Bureau • BEA
--	---	--

Data processing and analysis

It is spatial analysis-based research utilizing R. So, hardware with R software is a prerequisite for the study. To answer the questions, I will require tidyverse, dplyr, tmap, SpatialKDE, ggplot2, and leaflet library packages to create maps, graphs, and do buffer, intersection, density modeling, and raster analysis.

After collecting all the datasets from online sources, they have been added to the R environment for further cleaning and processing. To exclude Alaska and Hawaii from the study area, the dplyr library has been utilized. Using different functions of this library, such as filter, select, group_by, summarize, and mutate, multiple tables and data frames have been produced. Then ggplot is used to create charts and graphs, representing the outcome. For dynamic mapping, I applied the Leaflet function to create an interactive map of the distribution of EVCS. To understand the density, I operated SpatialKDE. To visualize the regional variation, tmap has been used to develop statistical maps.

Result

EV Adoption and Infrastructure Development

From 2016 to 2025, the registration of EVs, irrespective of regions, has increased, ranging from 4 million to 12 million. If we break it down into the types of EVs, it can be shown that the adoption rate of HEV, especially the PHEV, is dramatically higher than that of BEV (Fig. 1). One reason could be that BEV comes with only battery power, but HEV or PHEV allows users to use the car both as an electric-powered vehicle and a gasoline-powered vehicle. This flexibility reduces the anxiety of a user while making trips.

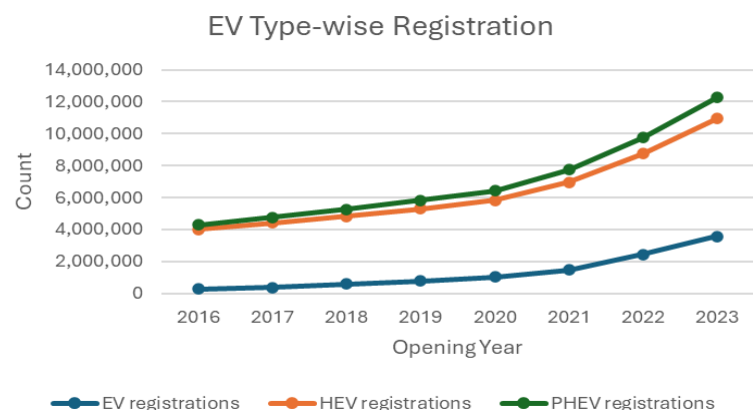


Fig. 1- EV Registration from 2016 to 2025

On the other hand, if we try to understand the development pattern of EV charging stations along with the registration, it is clear from the graph that the pattern is the same in the establishment trend, except that from 2022 to 2025 experiences a dramatic fall (Fig. 2). But a point to be noted is that that we are in the middle of 2025, which implies that the count is not final yet. Also, each year, the number of public charging stations is exponentially higher than that of private ones.

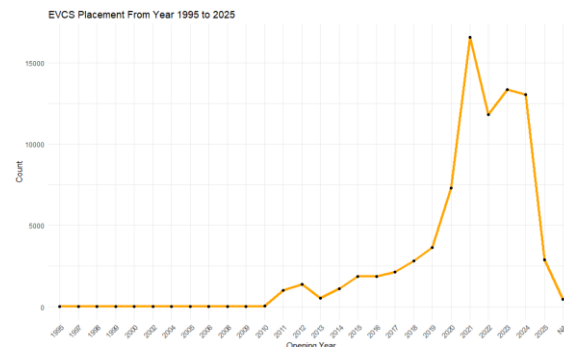
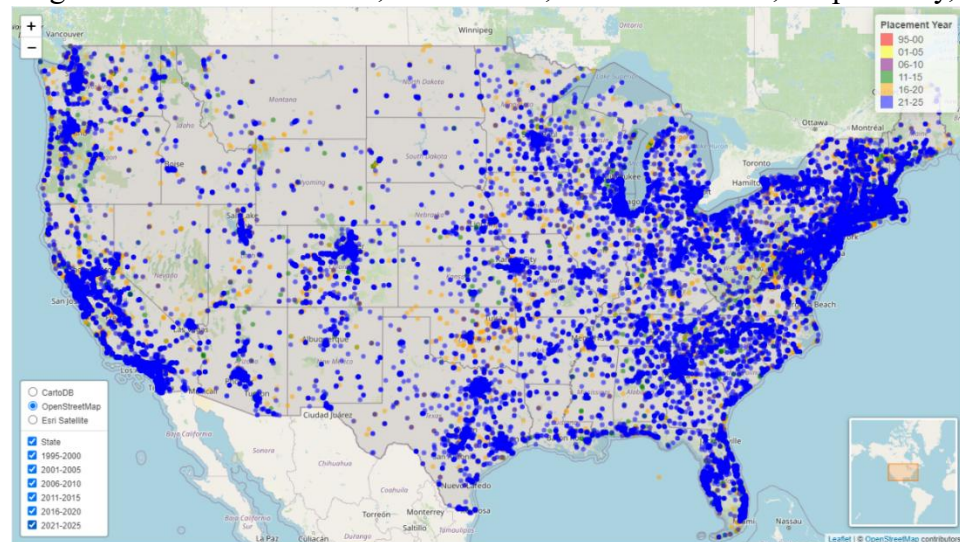


Fig. 2- Establishment of EVCS from 1995 to 2025

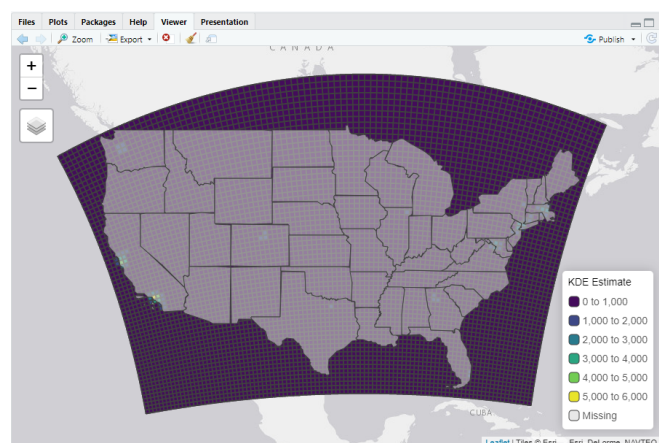
EVCS Density and Distribution

The map illustrates the distribution of EVCS throughout the United States, with color representing the establishment year of those stations. The most visible colors are blue, yellow, and green, representing the cohort 2021–2025, 2016–2020, and 2011–2015, respectively, which implies that significant numbers of development has been made between 2011 and 2021. A study concluded that government investment in this sector accelerated the development of EVCS.



Map 1: Establishment of EVCS over time

Though I initially failed to create a proper kernel density model with a satisfactory cell size and bandwidth, a map was eventually produced after numerous failed trials with larger cell sizes and bandwidths. The resulting map represents the concentration of EVCS in an area. If we look carefully at Map 2, some prominent color variations are noticed in California and the East Coast, indicating higher density in those regions.



Map 2: Kernel Density Model of EVCS distribution

EVCS with Charger Type and Charging Time

Table 2: EV Charger type with specifications

	Level 1	Level 2	DCFC (Level 3)
Locations	Home	Home, Workspace, Public	Public
Voltage	120 V AC	208 - 240 V AC	400 V - 1000 V DC
Est. Range: an hour of Charging	2 - 5 miles	10 - 20 miles	180 - 240 miles
Cost	\$1000 to \$5000 one-time installation cost, later electricity cost	\$1 to \$5 for an hour of charging	\$10 to \$30 per charge
BEV	40- 50+ hours	4-10 hours	20min- 1hour
PHEV	5-6 hours	1-2 hours	Not compatible

Table 2, representing the charger type with other detailing, helps us to understand the factors that can affect the charger choice of a user based on the charging time, cost, and their type of EV. It not only influences the choice of trips a user but also guides the development (Fig. 3).

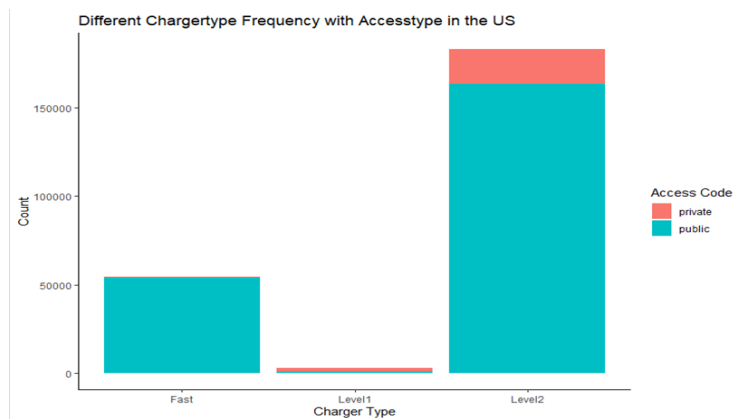
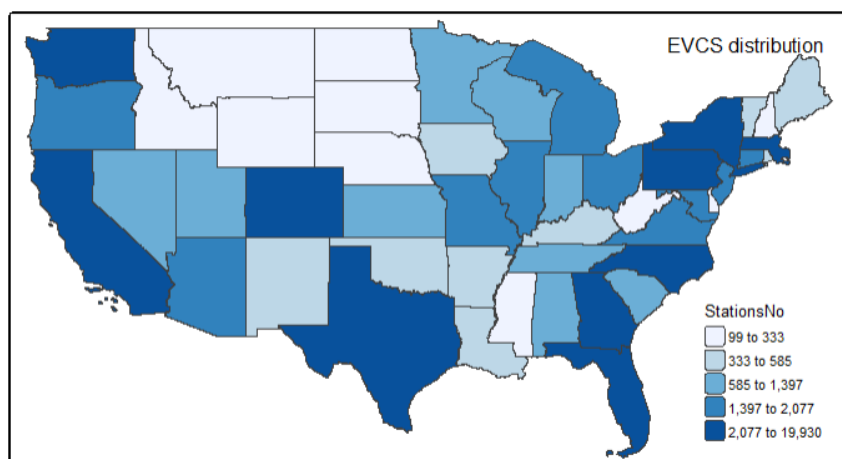


Fig.3: Available charger types with accessibility

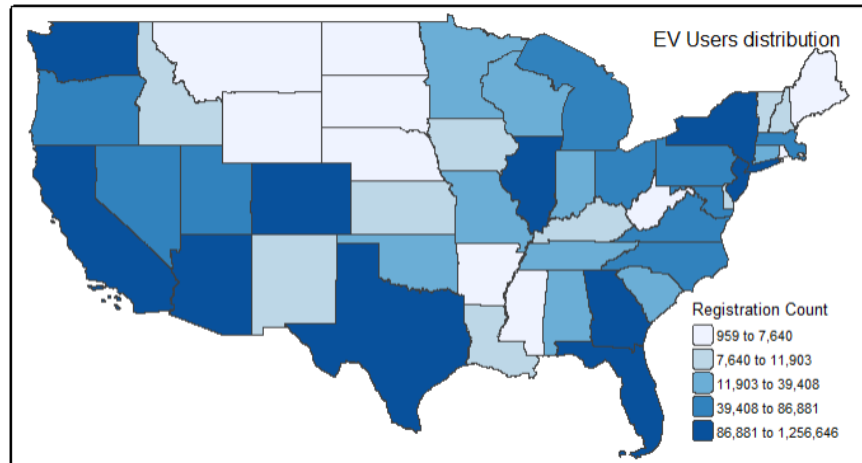
EVCS Regional Variation

In this part of the analysis, I examined the relationship between the distribution of EVCS and key demographic and economic indicators, including population density, median income, and each state's GDP contribution. The goal was to understand the trend of EV adoption and how EV infrastructure development aligns with the demographic conditions of different



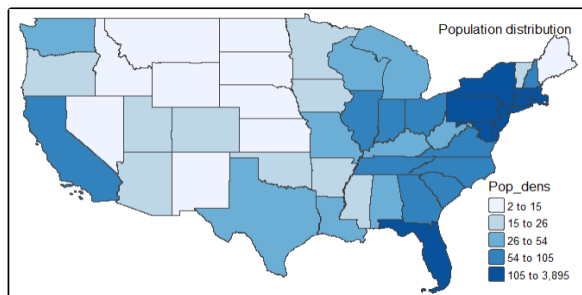
Map 3: State-wise total number of EVCS

states. As shown in Maps 3 and 4, the geographic patterns of EV adoption, represented by EV registration counts, and the total number of EVCS appear to follow a similar distribution, indicating a strong correlation between infrastructure presence and EV usage.

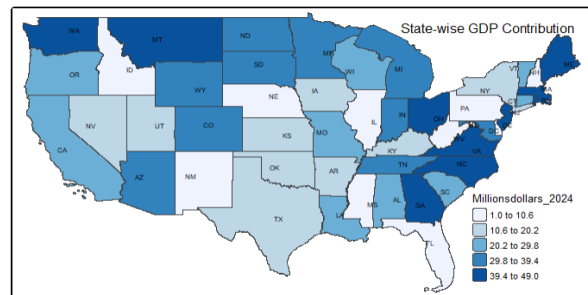


Map 4: State-wise EV registration count

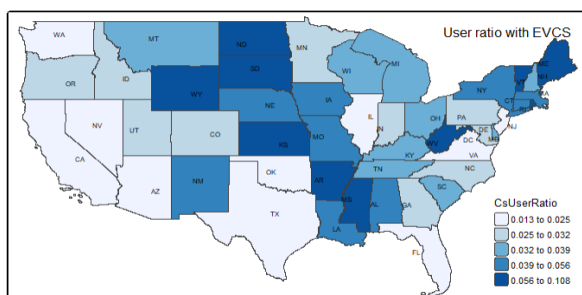
To compare this pattern with other demographic data, I generated a variable using the equation: $CSUserRatio = \text{Number of EVCS} / \text{EV Registration Count}$, and created a map, incorporating the values of the ratio (Map 7).



Map 5: State-wise population distribution



Map 6: GDP contribution of states



Map 7: State-wise ratio of EV users and



Map 8: Median income of States

However, when this ratio was compared with demographic and economic indicators (Maps 5, 6, 7, and 8), it became evident that the distribution is highly uneven. The CSUserRatio does not consistently align with population size, income levels, or state GDP, suggesting that EVCS deployment may not always reflect demographic needs or economic capacity.

Discussion

I had to deal with several challenges to carry out the whole work. Some of them are working with incompatible datasets, long waiting times due to file size, wrong selection of the extent of the study area, and insufficient remaining space in the environment of R. Additionally, I planned to work with road features and raster dataset, for which I prepared my dataset in R, but due to the file size, I could not generate the final output. Although there were lots of drawbacks in the process, I got a lot of experience, which will help to implement the same work differently. For example, I will select a smaller study area and work on a specific topic with more detail. I will organize the data more efficiently and be prepared beforehand about the file size and space. My initial idea was to work on the thesis questions, which I failed, but this failure came with lots of background knowledge and ideas for methodology development. However, I fully succeeded in utilizing R to produce my final maps and graphs, which were the prime requirements of the project.

Conclusion

This project has laid a solid foundation for my upcoming thesis by successfully exploring the distribution and density of EVCS in the United States, along with the adoption trends of electric vehicles over time. Despite initial challenges, the use of R proved effective in producing the required maps and visualizations. The findings clearly show regional variations, user preferences in EV types, and charging systems. This work will directly inform the next phase of my research, which will focus on EVCS placement along highways and optimal routing for long-distance EV travel. Ultimately, this work contributes meaningful data and analysis that can support policymakers and developers in addressing current gaps and planning for future EV infrastructure needs.

References

- Ahmad, F., Iqbal, A., Ashraf, I., Marzband, M., & Khan, I. (2022). Optimal location of electric vehicle charging station and its impact on distribution network: A review. *Energy Reports* 8, 2314–2333. Doi: 10.1016/j.egy.2022.01.180
- Ballinger, B., Stringer, M., Schmeda-Lopez, D. R., Kefford, B., Parkinson, B., Greig, C., & Smart, S. (2019). The vulnerability of electric vehicle deployment to critical mineral supply. *Applied Energy*, 255, 113844. DOI: 10.1016/j.apenergy.2019.11384
- Deb, S., Tammi, K., Kalita, K., & Mahanta, P. (2018). Review of recent trends in charging infrastructure planning for electric vehicles. *Wiley Interdisciplinary Reviews: Energy and Environment*, e306. DOI:10.1002/wene.306
- Ehsani, M., Longo, S., Gao, Y. & Ebrahimi, K. (2018) *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design*. CRC Press, Boca Raton.
- Erbas, M., Kabak, M., Ozceylan, E., & Çetinkaya, C. (2018). Optimal siting of electric vehicle charging stations: A GIS-based fuzzy Multi-Criteria Decision Analysis. *Energy. Volume 163*, Pages 1017-1031. DOI: 10.1016/j.energy.2018.08.140
- Garratt, E. (2024). *Why are electric cars more expensive?* The Electric Car Scheme. <https://www.electriccarscheme.com/blog/why-are-electric-cars-more-expensive#:~:text=The%20technology%20is%20more%20complex,and%20data%20is%20from%20SMMT>.
- GEOTAB (2025). Understanding EV range anxiety and tips on overcoming it. *EVs and Sustainability*. <https://www.geotab.com/blog/range-anxiety/>
- Hamon, M. (n.d.). *Electric Vehicle Charging Systems*. Pure Power Engineering. <http://purepower.com/blog/electric-vehicle-charging-systems>
- Javed, K. (2023). *EV - Pros and Cons of Fast Charging*. LinkedIn. <https://www.linkedin.com/pulse/fast-charging-javed-khan-ya3uf/>
- Mukherjee, J. C., & Gupta, A. (2015). A review of charge scheduling of electric vehicles in smart grid. *IEEE Systems Journal*, 9(4), 1541–1553.
- Qmerit. (November 18, 2024). *Is It Cheaper to Charge an EV at Home or in Public?* Comparing Long-Term Cost Analysis of EV Home Charging vs. Public Charging. Retrieved from <https://qmerit.com/blog/comparing-long-term-cost-analysis-of-ev-home-charging-vs-public-charging/>
- RMI. (2018). Retrieved from <https://www.rmi.org/wp-content/uploads/2017/10/RMI-From-Gas-To-Grid.pdf>
- Soczówka, P., Lasota, M., Franke, P., & Zochowska, R. (2024). Method of Determining New Locations for Electric Vehicle Charging Stations Using GIS Tools. *Energies* 2024, 17, 45-46. DOI: 10.3390/en17184546

- Sortomme, E., & El-Sharkawi, M. A. (2011). Optimal charging strategies for unidirectional vehicle-to-grid. *IEEE Transactions on Smart Grid*, 2(1), 131–138.
- Shepero, M., & Munkhammar, J. (2018). Spatial Markov chain model for electric vehicle charging in cities using geographical information system (GIS) data. *Applied Energy* 231 (2018), 1089–1099. DOI: 10.1016/j.apenergy.2018.09.175
- The Times of India. (2017). *NTPC forays into Electric Vehicle charging stations*. <http://timesofindia.indiatimes.com/business/india-business/ntpc-forays-into-electric-vehicles-charging-stations/articleshow/58975961.cms>
- U.S. Department of Energy (2023). *At a Glance: Electric Vehicles*. Office of Energy Efficiency and Renewable Energy. https://afdc.energy.gov/files/u/publication/electric-drive_vehicles.pdf
- U.S. Department of Energy (n.d.). *Electric Vehicle Benefits and Considerations*. Alternative Fuels Data Center. <https://afdc.energy.gov/fuels/electricity-benefits>
- U.S. Department of Energy (n.d.). *Electric Vehicle Charging Station Locations*. <https://afdc.energy.gov/fuels/electricity-locations#/find/nearest?fuel=ELEC>
- U.S. Department of Transportation. (2025). *Charger Types and Speeds*. <https://www.transportation.gov/rural/ev/toolkit/ev-basics/charging-speeds>
- U.S. Department of Transportation. (2016). *Transportation Toolkit*. <https://www.transportation.gov/policy-initiatives/leadership-academy/transportation-toolkit>
- U.S. Energy Information Administration. (2024). *Use of Energy Explained: Energy use for transportation*. <https://www.eia.gov/energyexplained/use-of-energy/transportation-in-depth.php>.
- Yong, J. Y., Ramachandaramurthy, V. K., Tan, K. M. & Mithulananthan, N. (2015). A review on the state-of-the-art technologies of electric vehicles, its impacts and prospects. *Renewable and Sustainable Energy Reviews* 49 (2015), 365–385. DOI: 10.1016/j.rser.2015.04.130
- Zhang, Y., & Iman, K. (2017). A multi-factor GIS method to identify optimal geographic locations for electric vehicle (EV) charging stations. *International Cartographic Association*, 1. DOI: 10.5194/ica-proc-1-127-2017